HOW CAN I WRITE THE RESEARCH PROPOSAL

SHARIFUL ISLAM MOZUMDAR

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CONTENTS

PREFACE

Writing a research proposal is the first step of every research project. It helps you plan what you want to study, why it is important, and how you will do it. But for many students, writing a proposal feels difficult and confusing. This book—"How Can I Write the Research Proposal"—will make that process easy. It is written for students and new researchers who want to learn how to write a good and simple proposal.

In this book, you will learn how to choose a research topic, find the research gap, write your objectives and methods, and prepare your proposal for submission. Each chapter uses simple English with examples and tips that you can follow easily. The goal of this book is to help you think like a researcher and explain your ideas clearly.

I hope this book will guide you step by step and give you confidence to write your own research proposal successfully.

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ABOUT THIS BOOK

This book is written to help students and new researchers learn how to write a research proposal step by step. It explains every part of a proposal in easy language so that anyone can understand and follow it. You will learn how to choose a topic, find the research problem, write your objectives, plan your methods, and clearly organize your proposal. Each chapter gives short examples, helpful tips, and simple templates you can use for your own work. You do not need to be an expert to use this book. It is made for beginners who want to learn and improve their writing skills. You can read the whole book from start to end, or you can open any chapter when you need help.

I hope it will guide you at every step—from the first idea to the final proposal submission. This book is your friend in the research journey—simple, clear, and practical.

HOW TO USE THIS GUIDE

This book is made to help you learn step by step. You don't have to read everything at once. Start with the chapter that you need most. If you are new to research, begin with Chapter 1 to understand what a research proposal is. If you already have a topic, go to the chapters about literature review or methodology. Each chapter is short and easy to read. You will find examples, tips, and simple steps to follow.

Keep a notebook beside you and write your own ideas as you read. You can also use this book when you start writing your real proposal. Remember—learning to write a research proposal takes time and practice. Read slowly, think carefully, and use this guide as your helper in the journey

-**Shariful Islam Mozumdar**

ACKNOWLEDGEMENTS

I would like to thank everyone who helped me in writing this book. First, I thank my teachers and mentors for their kind guidance and support. Their advice gave me the strength and ideas to write this guide for students and young researchers.I am also thankful to my friends and family for their love, patience, and encouragement during this work.

Finally, I thank all students and researchers whose questions and curiosity inspired me to write this book. I hope it helps you in your study and makes your research journey easier.

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CHAPTER 1:
UNDERSTANDING RESEARCH PROPOSALS

1.1 What Is a Research Proposal?

A research proposal is a short plan that explains what you want to study, why it is important, and how you will do it. It is like a map that shows the direction of your research. Your proposal tells others that you have a good idea, you understand the topic, and you know how to do the study. It helps your supervisor or funding body decide whether your plan is clear and possible.

In simple words, a research proposal is a promise—you promise to do your study in a logical and scientific way.

1.2 Purpose and Importance of a Research Proposal

A research proposal is important because it:

- helps you plan your research step by step,
- shows your understanding of the problem,
- convinces others that your study is worth doing, and
- helps you get approval or funding.

Without a proposal, your research may lose focus or direction. A clear proposal saves time, effort, and confusion later.

1.3 Difference Between a Proposal and a Research Paper

A research proposal is written before you start the actual study. It explains what you plan to do.

A research paper is written after you finish the study. It explains what you did and what you found.

Point	Research Proposal	Research Paper
Time	Written before research	Written after research
Purpose	To plan the study	To report results
Focus	What you will do	What you did
Audience	Supervisor or funding body	Scientific readers or journal

So, the proposal is the plan, and the research paper is the result.

1.4 Common Myths About Research Proposals

Many students feel afraid to write a research proposal because they believe some wrong ideas.

Here are a few common myths:

Wrong: “You need to know everything before you start.”

Right: No, you learn while doing research.

Wrong: “Only experts can write a proposal.”

Right: Anyone can learn to write with practice and guidance.

Wrong: “A long proposal means a better proposal.”

Right: Quality and clarity are more important than length.

Remember—a research proposal is not about being perfect. It is about showing that you have a clear, realistic, and interesting plan.

CHAPTER 2: CHOOSING A RESEARCH TOPIC

2.1 How to Find a Research Problem

Every research starts with a problem or a question. A good research problem is something that is not fully solved or understood yet. To find a research problem:

- Read books, journals, and articles in your subject area.
- Observe real-life issues in your community or field.
- Talk to teachers, experts, or classmates.
- Think about what interests you most.

A strong research problem should be clear, specific, and researchable—something you can study within your time and resources.

2.2 Identifying the Research Gap

A research gap means something that previous studies have not explained well or have not studied enough.
To find the gap:

- Read the literature carefully.
- Notice what questions are still unanswered.
- Find where researchers say, “more study is needed.”

Your proposal should show that you understand what has been done before and what is still missing — and that your study will help fill that gap.

2.3 Formulating Research Questions and Objectives

Once you know your topic and gap, you need to write research questions and objectives.
They show what you want to discover and how you will do it.
Example:

- Research Question: How can local microorganisms remove ammonia from aquaculture wastewater?

- Research Objective: To test the efficiency of indigenous microorganisms for ammonia removal in lab and field conditions.

Tips:

- Keep them short and clear.
- Use action verbs like identify, evaluate, develop, compare, and analyze.
- Make sure each question matches one objective.

Good questions and objectives will guide your whole proposal and keep your study focused.

Briefly:

Broad Research Area

↓

Specific Research Topic

↓

[Reading Literature

↓

Identify Research Gap

↓

Write the general research problem.

↓

Formulate Research Questions

↓

Write Research Objectives

↓

Each Question ↔ One Objective

CHAPTER 3: REVIEWING THE LITERATURE

3.1 What Is a Literature Review?

A literature review means reading and summarizing what other researchers have already written about your topic. It helps you understand what is known and what is still unknown. The main purpose of a literature review is to:

- learn from previous studies,
- find the research gap, and
- show that your study is needed.

In simple words, a literature review tells the story of what others have discovered before you—and where your work fits in.

3.2 How to Search and Organize Literature

To write a good literature review, you need to find, read, and organize information from reliable sources.

Step 1: Search for Information

- Use academic databases like Google Scholar, Scopus, PubMed, or ResearchGate.
- Use keywords related to your topic (e.g., ammonia removal, indigenous microorganisms, aquaculture wastewater).
- Read the title, abstract, and conclusion first—to see if it is useful.

Step 2: Organize Your Sources

- Create a table or note system to record:
    - Author and year
    - Title and journal name
    - Main findings
    - Gaps or limitations

Step 3: Summarize

- Write short summaries in your own words.
- Group similar studies together (e.g., by method or result).

This way, your literature review will be logical and easy to follow.

3.3 Writing a Critical Summary

A good literature review is not just a list of articles — it is a critical summary. That means you must analyze, compare, and comment on what you read.

Example:

Many studies have used biofilters for ammonia removal in aquaculture wastewater (Ali et al., 2020; Chen et al., 2022). However, most of these systems depend on imported microbial strains, which increases cost and reduces long-term stability.

Few studies have focused on using indigenous microorganisms for sustainable ammonia removal. This kind of writing shows that you understand both the strengths and weaknesses of previous research.

Tips for a Strong Literature Review

- Use recent and reliable sources (last 5–10 years).
- Always cite your sources properly (APA or Harvard style).
- Write in your own words—avoid copy-paste.
- Move from general ideas to specific studies.
- End the review by highlighting the research gap that your study will fill.

CHAPTER 4:
RESEARCH DESIGN AND METHODOLOGY

4.1 Research Approach (Qualitative, Quantitative, Mixed)

There are three main types of research approaches:

1. Qualitative Research:
   - Focuses on understanding people's experiences, opinions, or behaviors.
   - Uses interviews, observations, or open-ended questions.
   - Example: Interviewing farmers about the challenges of fish farming.

2. Quantitative Research:
   - Focuses on numbers, measurements, and statistical analysis.
   - Uses experiments, surveys, or lab tests.
   - Example: Measuring ammonia levels in aquaculture wastewater.

3. Mixed-Methods Research:
   - Combines both qualitative and quantitative approaches.
   - Example: Testing ammonia removal (quantitative) and interviewing operators about system performance (qualitative).

Choose the approach that best fits your research question and objectives.

CHAPTER 4:
RESEARCH DESIGN AND METHODOLOGY

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   - Example: Measuring ammonia levels in aquaculture wastewater.

3. Mixed-Methods Research:
   - Combines both qualitative and quantitative approaches.
   - Example: Testing ammonia removal (quantitative) and interviewing operators about system performance (qualitative).

Choose the approach that best fits your research question and objectives.

4.2 Sampling and Data Collection

Sampling means selecting a part of the population or area to study. You can't always study everything, so you choose a smaller, representative group.

Types of Sampling:

- Random Sampling: Everyone has an equal chance to be selected.
- Purposive Sampling: You choose samples based on specific reasons.
- Stratified Sampling: You divide the population into groups and select samples from each.

Data Collection Methods:

- Primary Data: Data you collect yourself (e.g. lab experiment, field observation, survey).
- Secondary Data: Data already collected by others (e.g., books, journals, reports).

Example:
Water samples will be collected from three different aquaculture ponds. Ammonia concentration will be measured weekly using standard laboratory methods.
Always explain how, where, and when you will collect your data.

4.3 Tools, Instruments, and Ethical Considerations

Tools and Instruments:
Mention the equipment, materials, and methods you will use in your study.
Example:

- pH meter, spectrophotometer, dissolved oxygen meter
- Microbial culture media and incubator
- Questionnaire or interview guide (for surveys)

Describe how you will use each tool to collect accurate and reliable data.

Ethical Considerations:

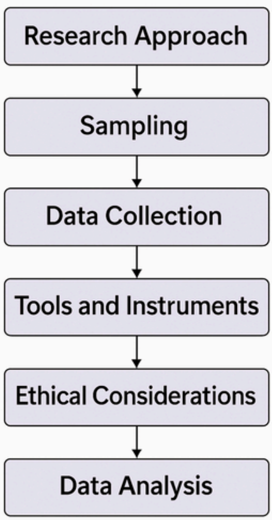
All research must follow ethical rules to ensure honesty and respect for others.

- Do not harm people, animals, or the environment.
- Get permission before collecting data.
- Give credit to all authors and sources (avoid plagiarism).
- Be truthful in recording and reporting your results.

Ethics make your research trustworthy and professional.

Summary

Your research design and methodology show how your study will be carried out. It connects your objectives with real actions—from choosing your approach to collecting data to analyzing it ethically and carefully.



Steps in Research Methodology

CHAPTER 5:
PROPOSAL STRUCTURE AND WRITING STEPS

A research proposal follows a clear and logical structure. Each part helps explain what you plan to study, why it is important, and how you will do it. Below are the main parts of a standard proposal, along with step-by-step instructions for writing each section.

Standard Structure of a Research Proposal

- Title Page
- Abstract or Executive Summary (if needed)
- Introduction / Background of the Study
- Problem Statement
- Research Questions/Hypotheses
- Research Objectives
- Literature Review
- Research Methodology
- Expected Results / Outcomes
- Research Timeline (Work Plan)
- Budget and Justification (if required)
- Limitations of the Study (optional but useful)
- References/Bibliography Citation Style
- Appendices (if needed)

“A good research proposal tells what, why, and how clearly—through a logical structure.” It demonstrates the researcher’s clarity of thought, depth of understanding, and scientific planning. A strong proposal is not just an idea—it is a roadmap that convinces others your study is worth doing.

5.1 How to Write the Title Page

The title page is the very first part of a research proposal and serves as the formal introduction of your study. It gives readers a clear idea of what your research is about and who you are as a researcher. Although it is only one page long, it plays an important role in creating the first impression. A neat and well-prepared title page reflects your professionalism and attention to detail.

A good title page should be simple, clear, and properly organized. It usually includes the essential components of the title page.

- Title of the Research Proposal
- Researcher's Full Name
- Department and Institution
- Supervisor's Name and Designation
- Degree and Purpose of Submission
- Month and Year of Submission
- (Optional) Institutional Logo

All information should be center-aligned and neatly arranged.

When writing the research title, make sure it clearly expresses what you want to study, where the study applies, and how it will be carried out. The title should be short but meaningful—usually between 15 to 20 words. Avoid unnecessary words like "A study on" or "An investigation of" unless required. Instead, use direct and descriptive phrases that show the main focus of your research.

Key Elements of a Good Research Title

A good research title is not just short and attractive—it must clearly reflect the core design of your study. Every effective title should include three essential elements:

- The Subject
- The Treatment and
- The Parameter.

Together, these show what is being studied, how it is being treated, and what is being measured.

A. Subject (What you are studying): The subject is the main focus or material of your research—the topic, organism, or system you are studying. Example: Aquaculture wastewater, tomato plants, etc., to be the subject.

B. Treatment (What is being applied or tested): The treatment refers to the method, condition, or factor you are applying or testing to bring change. Example: By immobilized microorganisms, under organic fertilizers, using blended learning, etc., to be the treatment.

C. Parameter (What you are measuring): The parameter is the variable or response you are observing to evaluate the effect of the treatment. Example: For ammonia removal efficiency, on growth performance, to assess motivation, etc., to be the parameter.

As an example:
Subject: Aquaculture wastewater
Treatment: Immobilized indigenous microorganisms
Parameter: Ammonia removal efficiency.

The final title could be "Development of Efficient Ammonia Removal from Aquaculture Wastewater by Immobilized Indigenous Microorganisms."

This title is clear, specific, and scientifically structured—showing exactly what is being studied, how, and for what purpose.

5.2 Abstract: How to Write It

The abstract is the summary of a research proposal. It gives readers a quick but complete picture of the study—explaining what will be done, why it matters, and how it will be carried out. Although it is short, usually 150 to 250 words, the abstract is one of the most important parts of a proposal because it forms the first academic impression of your work.

Purpose of the Abstract: The purpose of the abstract is to summarize your proposal in a compact form. It highlights the research problem, objectives, methods, and expected outcomes, giving a quick overview of the study. Reviewers, funding bodies, and readers often decide whether to read the full proposal based on the quality of the abstract. Therefore, your abstract should be both concise and convincing.

Structure of a Good Abstract: A well-written abstract usually follows a logical structure consisting of five short but connected parts:

- Background: One or two sentences introducing the research context or problem.
- Objective: A clear statement of what the research aims to achieve.

- Methodology: The main approach, design, or materials to be used in the study.
- Expected Results: The anticipated findings or outcomes.
- Significance: The importance or practical value of the research.

Each part should be written in simple, direct sentences and flow smoothly in a single paragraph. Avoid bullet points or subheadings.

Writing Style:

- Write in the present or future tense, depending on the situation.
- Avoid unnecessary details, tables, figures, and references.
- Do not include abbreviations unless they are very common.
- Use active voice and straightforward language.
- Keep the tone academic but easy to read.

A good abstract should be short enough to read quickly, yet detailed enough to reflect the entire study.

Keywords: After the abstract, include 5–7 keywords that represent the main concepts of the study. These should be derived from the subject, treatment, and parameter of your title. They help readers and databases identify your research area easily.

Common Mistakes to Avoid

- Writing the abstract before finishing the proposal.
- Adding too much background or theory.
- Using vague phrases like "This study is about…" instead of direct statements.
- Exceeding the word limit or omitting key details.

5.3 Introduction—How to Write It

The introduction is the first and most important part of a research proposal. It gives readers the background and reason for your study—explaining why the research is needed, what problem it solves, and how it connects to existing knowledge. A good introduction catches the reader's interest and clearly shows the purpose of your work. It is not just general information—it is the base that supports your whole research proposal.

Purpose of the Introduction: The main purpose of the introduction is to guide the reader from the general topic area to the specific focus of your study.

It answers the following key questions:

- What is the background of the research?
- What is the problem or gap in existing knowledge?
- Why is the study important or relevant?
- What are the objectives that this research aims to achieve?

Structure of a Good Introduction: A good introduction follows a logical flow—beginning with broad information and gradually narrowing down to your specific study. It generally consists of four essential parts:

- **General Background:** Start with a brief discussion of the broader field or issue related to your topic. Provide general information, trends, or facts that help the reader understand the overall context. You may use statistics or global reports to make the introduction engaging.

Example: "Aquaculture has become one of the fastest-growing food production sectors worldwide, providing more than half of the global fish supply (FAO, 2024)."

- **Problem Context and Gap:** After setting the background, move to the specific problem that your study will address. Explain what previous studies have done and where the gap still exists. Use citations to show the progression of research and justify your topic.

Example: "Although several studies have investigated biological nitrogen removal systems (Chen et al., 2021; Lee & Park, 2023), limited research has focused on using local microorganisms immobilized on natural substrates in tropical aquaculture systems."

- **Rationale or Significance:** Here, describe why your study is important. Explain how it contributes to solving real-world problems, advancing scientific knowledge, or supporting sustainable development.

Example: "Developing an efficient, low-cost ammonia removal system using indigenous microbes can improve aquaculture water quality and promote environmentally friendly production practices."

- **Research Focus and Objectives:** Finally, narrow down to your specific research focus. State the main goal, aim, or hypothesis that your proposal will address. This last paragraph connects directly to the next section (Problem Statement or Objectives).

Example:
"This study aims to develop a biological treatment process for ammonia removal from aquaculture wastewater using immobilized indigenous microorganisms and to evaluate its performance under controlled laboratory conditions."

Writing Style for the Introduction: To make your introduction effective and readable:

- Start broad and move toward the specific problem.
- Maintain a logical and coherent flow between paragraphs.
- Use simple, formal, and factual language.
- Avoid unnecessary theory, long quotations, or too much technical data.
- Always support your statements with credible citations.
- Do not include detailed methods or results—those belong to later sections.

Using Citations in the Introduction: Citations are essential in the introduction or background. They provide scientific evidence for your statements, demonstrate your literature knowledge, and help identify the research gap. Without proper citations, your introduction becomes opinion-based; with citations, it becomes academically strong. How Many Citations Should Be Used?

- For the Level of undergraduate proposal, the recommended number of citations is 5-10
- For the level of a master's proposal, the recommended number of citations is 10–15.
- For the level of a PhD or grant proposal, the recommended number of citations is 15–25.

Use recent, peer-reviewed sources (preferably within the last 5–10 years).

Where to Use Citations:

- General Background: 2–3 citations for global/local context.
- Problem Context: 4–6 citations to show what is already known.
- Research Gap: 2–3 citations to identify what is missing.

Linking Citations with Subject, Treatment, and Parameter

When writing the background, keep your subject, treatment, and parameter in mind:

Element	Focus of Citation	Example
Subject	Field or system under study	"Aquaculture wastewater contains high ammonia levels (FAO, 2024)."
Treatment	Applied method or technology	"Immobilized microbial systems enhance biofiltration efficiency (Wang et al., 2021)."
Parameter	Variable being measured	"Ammonia removal efficiency depends on microbial density and carrier type (Li et al., 2022)."

This keeps your introduction structured and scientifically relevant.

Common Mistakes to Avoid

- Starting with a very narrow topic without a broader context.
- Copying background text from previous papers.
- Including detailed methods or results.
- Using too few or outdated citations.
- Failing to identify a clear research gap.

5.4 Problem Statement

The problem statement is one of the most critical parts of a research proposal. It clearly defines what problem your study aims to solve and why it needs to be studied. A well-written problem statement helps readers understand the research gap and justifies the need for your investigation. In simple terms, it tells what is wrong, what is missing, and what you will do to address it.

Purpose of the Problem Statement: The main purpose of the problem statement is to:

- Identify the existing issue or gap in knowledge, practice, or understanding.
- Explain why this issue is important and needs research.
- Show how your study will help to fill that gap or provide a solution.

It connects the background or introduction with your objectives, giving a clear direction to the entire proposal.

Structure of a Good Problem Statement: A strong problem statement usually includes three logical parts:

- The Context or Background: Briefly introduce the general topic and existing condition.
- The Gap or Problem: Point out what is missing, unknown, or inefficient in current knowledge or practice.
- The Need for the Study (Justification): Explain why it is important to study this problem now and how your research will help.

Relation to Subject, Treatment, and Parameter: When writing the problem statement, always align it with the three elements of your research title:

Element	Focus in Problem Statement	Example
Subject	The area or system facing the problem	Aquaculture wastewater contains high ammonia levels.
Treatment	The existing gap in the current treatment or method	Chemical methods are costly and unsustainable.
Parameter	The measurable issue that needs improvement	Low efficiency in ammonia removal or poor water quality.

Thus, your problem statement should naturally point toward what will be improved, how, and why it matters.

Writing Tips

- Keep the problem statement short—around 150–200 words.
- Be specific—avoid vague phrases like "this topic is interesting."
- Support your statements with 1–2 citations if needed.
- Use a formal and objective tone.
- The last sentence should lead directly to your research objectives.

Common Mistakes to Avoid

- Writing a general background instead of a real problem.
- Mentioning multiple unrelated problems.
- Copying problem statements from other studies.
- Failing to connect it with your subject, treatment, and parameter.

5.5 Research Questions/Hypotheses

After identifying the research problem, the next step is to translate it into specific questions or testable hypotheses. These form the core framework of your study—they guide data collection, analysis, and interpretation. A good research question or hypothesis ensures that your study remains focused, measurable, and scientifically meaningful.

Purpose: The main purposes of research questions or hypotheses are to:

- Define the scope and direction of your research.
- Transform a broad problem into clear, answerable inquiries, and
- Provide a basis for testing and concluding.

In simple words, the research question asks, "What do I want to find out?" and the hypothesis predicts, "What do I expect to happen?"

A. Research Questions: Research questions are used in exploratory or descriptive studies, where the goal is to investigate relationships, patterns, or behaviors rather than test predictions. Each question should directly relate to your objectives and be specific and focused. Characteristics of Good Research Questions.

- They are clear and concise.
- They can be answered through data collection and analysis.
- They are logically derived from the problem statement.
- They reflect the subject, treatment, and parameter of your study.

B. Hypotheses: A hypothesis is a formal statement that predicts the expected relationship between variables. It is mostly used in experimental or quantitative research where testing is required. Types of Hypotheses

Null Hypothesis (H_0):

- States that there is no significant relationship or difference between variables. Example: H_0 – Immobilized indigenous microorganisms have no significant effect on ammonia removal efficiency.

Alternative Hypothesis (H_1):

- States that there is a significant relationship or difference. Example: H_1 – Immobilized indigenous microorganisms significantly improve ammonia removal efficiency in aquaculture wastewater.

Writing Guidelines

- Align your questions/hypotheses with the objectives of the study.
- Use simple and precise language.
- Avoid overlapping or repetitive questions.
- Limit to 3–5 main research questions or hypotheses.
- Ensure they are measurable, testable, and related to your subject, treatment, and parameter.

Common Mistakes to Avoid

- Making research questions too broad or vague.
- Writing hypotheses that are not measurable.
- Including too many questions without a clear focus.
- Not aligning questions with objectives or the problem statement.

Research questions and hypotheses act as the compass of your study—they point the direction, set the limits, and define how your research will discover new knowledge.

5.6 Research Objectives

The Research Objectives section describes what your study intends to accomplish. It translates the research problem into clear, specific, and achievable goals. While the problem statement identifies what is wrong and the research questions ask what you want to find out, the objectives explain what you will do to find the answers. In other words, this section tells the reader the purpose, scope, and expected direction of your research.

Purpose of Research Objectives: Research objectives serve several key purposes:

- To define the aim and focus of your study clearly.
- To guide the research design, data collection, and analysis.
- To connect your problem statement with the expected outcomes.
- To help reviewers understand the feasibility and significance of your proposal.

Without clear objectives, a proposal looks directionless-Objectives are the compass that keeps your research on the right path.

Types of Research Objectives: Research objectives are generally divided into two main types:

- **General Objective** (Overall Goal) The general objective expresses the broad aim or ultimate purpose of your study. It summarizes the central focus in one sentence.

- **Specific Objectives:** The specific objectives break down the general goal into smaller, measurable tasks. Each specific objective should answer one key aspect of your research question and should be clear, actionable, and measurable. Example:

1. To isolate and enrich indigenous nitrifying microorganisms from aquaculture pond water and sediment.
2. To characterize the microbial community responsible for ammonia oxidation.
3. To immobilize the selected microorganisms on natural substrates such as coconut husk.
4. To evaluate the efficiency of immobilized microorganisms in ammonia removal under laboratory conditions.
5. To compare the treatment performance with conventional biofiltration methods.

Each of these specific objectives supports the general aim and addresses one dimension of the subject, treatment, and parameter relationship.

Writing Guidelines

- Write one general and 3–5 specific objectives.
- Use action verbs such as "to determine," to analyze," "to evaluate," "to develop," "to compare," and "to optimize."
- Keep each objective clear and measurable—avoid vague terms like "to study" or "to understand."
- Make sure your objectives can be realistically achieved within your time and resources.
- Arrange them in a logical order—usually following your research flow or methodology.

Relation with Subject, Treatment, and Parameter

When formulating objectives, ensure that each one relates to these three components:

Component	How It Relates to Objectives	Example
Subject	Focus on the area or system studied	"To study ammonia removal in aquaculture wastewater."
Treatment	Focus on the method or factor applied	"To apply immobilized indigenous microorganisms for treatment."
Parameter	Focus on what will be measured or evaluated	"To determine ammonia removal efficiency and water quality improvement."

This alignment makes your objectives scientific, relevant, and logically connected to your title.

Common Mistakes to Avoid

- Writing too many objectives (more than 6–7).
- Mixing different ideas in one objective.
- Using objectives that are not measurable or testable.
- Repeating the same content already covered in the research questions.
- Forgetting to align them with the problem statement.

5.7 Literature Review: How to Write It

The literature review is one of the most important parts of a research proposal. It provides the theoretical background and existing knowledge related to your study. By reviewing previous studies, you demonstrate what is already known, what is still unknown, and how your work will fill that gap.

In short, the literature review shows that your research is not starting from zero—it is built upon a foundation of earlier findings and scholarly evidence.

Purpose of the Literature Review: The main purposes of a literature review are:

- To summarize what has already been studied on your topic.
- To identify the research gap that your study will address.
- To justify the need and novelty of your work.
- To build a conceptual or theoretical framework for your research design.
- To demonstrate your knowledge and understanding of your field.

A strong literature review not only describes previous research—it analyzes, compares, and connects those studies with your own research focus.

Structure of a Literature Review: A good literature review follows a logical flow—from general concepts to specific studies directly related to your topic. It can be divided into four main parts:

1. **Introduction to the Topic**
    - Present the general background of your research area.
    - Define key terms and core concepts.
    - Explain the importance of studying this topic.
2. **Review of Previous Studies**
    - Summarize key findings from earlier research.
    - Compare and contrast different approaches, results, or conclusions.
    - Identify patterns, similarities, and inconsistencies.
3. **Research Gap Identification**
    - Clearly point out what is missing or unexplored in existing studies.
    - Highlight the limitations of previous works.
    - Show how your research will fill this gap or improve upon past methods.
4. **Link to Current Study**
    - Connect your topic with the reviewed literature.
    - Explain how your study builds on, extends, or differs from earlier research.

Writing Style Tips

- Use paraphrasing, not copy-paste—always write in your own words.
- Include citations for every key finding (APA or Harvard style).
- Use recent references (preferably from the last 5–10 years).
- Use transition words to maintain flow—similarly, however, moreover, and in contrast.
- Do not summarize each study separately; instead, synthesize ideas.
- End with a clear justification for why your research is needed.

Organizing the Literature Review: There are three common ways to organize a literature review:

Approach	Description	Example Use
Chronological	Arranges studies in order of publication (oldest to newest).	When showing the historical development of an idea.
Thematic (Topical)	Groups studies by themes, concepts, or variables.	When discussing different factors affecting ammonia removal.
Methodological	Compares methods and approaches used by different researchers.	When analyzing different wastewater treatment techniques.

Citation Guidelines:

- Use 10–20 high-quality references depending on proposal length.
- Prefer peer-reviewed journals, books, and reports over websites.
- Follow a consistent referencing style (e.g., APA 7th edition).

Common Mistakes to Avoid

- Writing long summaries of each study separately.
- Using outdated or non-academic sources.
- Forgetting to identify the research gap.
- Lacking logical flow or transitions between ideas.
- Copying content without paraphrasing and citation.

5.8 Research Methodology – How to Write It

The research methodology is the backbone of your entire proposal. It explains how you will carry out your study—the design, tools, procedures, and analysis you will use to achieve your research objectives. While the introduction tells why your research is important, the methodology tells how you will make it happen. A clear and well-structured methodology assures readers that your study is scientific, systematic, and feasible.

Purpose of the Methodology

The purpose of the methodology section is to:

- Describe the research design and overall approach.
- Explain the process of data collection and analysis.
- Show that your study can be replicated or verified.
- Ensure that your methods are appropriate for answering the research questions and achieving the objectives.

In short, this section tells reviewers, "What will you do, how will you do it, and why is this the best way to do it?"

Structure of a Research Methodology:

The structure of a research methodology provides a clear outline of how the study will be designed, conducted, and analyzed. It helps both the researcher and the reader understand the logical flow of the investigation —from defining the research type to ensuring ethical practices. A well-structured methodology answers three key questions:

(1) What will you do?

(2) How will you do it?

(3) Why is this the best way to do it?

Main Components of a Research Methodology

A complete methodology section typically includes eight essential components, arranged in a logical order:

1. Research Design: This describes the overall plan or approach you will use to achieve your objectives. It defines whether your research is quantitative, qualitative, or mixed-method, and specifies the type of study—such as experimental, descriptive, comparative, or exploratory.
Example: “This study will adopt an experimental research design to evaluate the effect of selected treatments on performance indicators.”

2. Study Area or Population: Here, you identify the location of the study and the target group or material you will focus on. In natural sciences, it may include sampling sites or organisms; in social sciences, it refers to the people or communities under investigation.
Example: “The study will be conducted among secondary school students in selected institutions.”

3. Sampling Technique and Sample Size: Explain how participants, samples, or data sources will be selected. Mention the sampling method (e.g., random, stratified, purposive, or convenience) and the number of samples or respondents, with justification.
Example: “A random sampling method will be used to select 150 respondents to ensure unbiased representation.”

4. Data Collection Methods: This section explains how information will be gathered to answer your research questions.

You may use surveys, interviews, laboratory experiments, observations, or document reviews. Describe the instruments (e.g., questionnaire, sensors, reagents) and the procedures used to ensure data accuracy and reliability.
Example: “Primary data will be collected using a structured questionnaire, while laboratory tests will provide quantitative measurements.”

5. Research Instruments or Materials: If your study requires specific tools or materials, list them clearly. This may include measuring devices, chemicals, or standardized instruments for social or physical research.
Example: “The study will use laboratory glassware, reagents, and digital pH meters for sample analysis.”

6. Data Analysis Techniques: Describe how the collected data will be processed and interpreted. For quantitative studies, mention the statistical software (SPSS, R, Excel) and analysis methods (mean, ANOVA, regression, correlation). For qualitative studies, describe techniques like thematic analysis or coding.
Example: “Quantitative data will be analyzed using descriptive and inferential statistics with SPSS software.”

7. Ethical Considerations: Every scientific study must maintain ethical standards. Mention how you will ensure informed consent, confidentiality, accuracy, and integrity. If necessary, state that ethical approval will be obtained from the appropriate committee.
Example: “Ethical clearance will be obtained before data collection, and participant confidentiality will be maintained throughout the study.”

8. Limitations of the Study (Optional but recommended): Acknowledge possible challenges such as limited sample size, time, or resources. and explain how you plan to minimize their effect on your results.
Example: "Although time limitations may restrict the number of observations, repeated sampling will be conducted to ensure data validity."

Summary of Methodology Structure

Component	Purpose	Key Question Answered
Research Design	Defines the overall plan	What approach will be used?
Study Area / Population	Identifies who or what is being studied	Where and on whom will the study focus?
Sampling	Selects participants or samples	How will the data sources be chosen?
Data Collection	Gathers necessary information	How will the data be obtained?
Instruments / Materials	Lists tools or resources used	What tools or materials are needed?
Data Analysis	Explains how data will be processed	How will the data be analyzed?
Ethics	Ensures integrity and safety	How will ethical issues be managed?
Limitations	Addresses constraints	What possible challenges may arise?

Writing Tips

- Use clear headings for each component.
- Maintain a logical flow—from design to data analysis.
- Write in the future tense when describing a proposed study.
- Be concise but detailed enough for replication.
- Justify each choice briefly (why you selected a particular method).

5.9 Expected Results / Outcomes

The Expected Results section describes what outcomes are likely to come from your research. It shows the reader what new knowledge, improvement, or solution your study will provide. These results must be realistic, measurable, and directly related to your research objectives.

Purpose: The purpose of this section is to:

- Show what the study will produce or prove.
- Indicate how the findings will add value to the existing knowledge.
- Help readers understand the possible benefits of your research.

How to Write Expected Results: Write a short paragraph (or 3–5 bullet points) describing:

- What you expect to find or develop.
- What measurable improvements or changes do you expect?
- How the results will help solve the problem or improve understanding.

Writing Tips

- Use clear headings for each component.
- Maintain a logical flow—from design to data analysis.
- Write in the future tense when describing a proposed study.
- Be concise but detailed enough for replication.
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5.10 Research Timeline (Work Plan)

The research timeline, or work plan, shows how your research activities will be carried out over a specific period of time. It helps to organize your work systematically and ensures that every step of the study can be completed within the available time. A well-prepared timeline reflects good planning, discipline, and feasibility of your research proposal.

Purpose of a Research Timeline:

The main purposes are:

- To show that your research can be completed within the given time.
- To organize activities in a logical and sequential order.
- To demonstrate proper time management and planning skills.

How to Write a Research Timeline

- Divide your research process into main stages or activities.
- Estimate how much time each activity will take.
- Present the plan in a table or Gantt chart format for clarity.
- Keep it realistic and balanced—not too tight or too long.

Common Research Activities to Include

- Title selection and proposal writing
- Literature review
- Research design and planning
- Data collection or experimentation
- Data analysis and interpretation
- Report or thesis writing
- Proofreading, submission, and presentation.

Example (6-Month Research Timeline)

Research Activities	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Topic selection and proposal writing	●●●					
Literature review	●●●●	●●				
Methodology design and pilot study		●●●				
Data collection / Experimentation		●●●●	●●●●			
Data analysis			●●●●	●●●		
Writing and editing of report				●●●●	●●●	
Final review and submission					●●●	●●●●

(Here "●" indicates work progress for each month)

Writing Tips

- Keep the timeline short and realistic—not over-ambitious.
- Match the schedule with your research objectives and methodology.
- Use a simple table or chart so that readers can easily understand your work plan.
- Mention the total duration of your study (e.g., 6 months, 1 year).
- Ensure that important tasks (data collection, analysis, and report writing) get enough time.

A good work plan is like a roadmap—it shows when and how each step of your research will happen. ensuring that your study stays on track from start to finish.

5.11 Budget and Justification (if required)

The Budget section presents a clear and realistic estimate of the total cost needed to carry out your research. It shows how the available funds will be used efficiently for various research activities such as data collection, materials, travel, and analysis. If required, a justification should be added to explain why each expense is necessary for achieving the research objectives. A well-prepared budget reflects that your proposal is practical, transparent, and financially feasible.

Purpose of the Budget

The main purposes of including a budget are

- To show the financial requirements of the research project.
- To ensure proper use of resources and prevent overspending.
- To demonstrate accountability and good planning to sponsors or institutions.
- To prove that the study is feasible within available funds.

Components of a Standard Research Budget

A typical research budget includes the following cost categories:

Budget Category	Description / Examples
A. Personnel / Research Assistant	Payment or allowance for research helpers, field workers, or data collectors.
B. Materials and Supplies	Chemicals, laboratory items, stationery, printing, paper, etc.
C. Equipment (if needed)	Instruments, sensors, or tools required for data collection or testing.
D. Travel and Field Expenses	Transportation, accommodation, or site visit costs.
E. Data Collection and Analysis	Survey costs, software licenses, or laboratory testing fees.
F. Documentation and Report Preparation	Typing, printing, binding, or editing of final report.
G. Contingency (optional)	Reserve fund (usually 5–10%) for unexpected costs.

Example of a Simple Budget Table

Item / Activity	Estimated Cost (USD)	Justification
Research Assistant (3 months)	300.00	For data collection and analysis support
Laboratory Materials & Chemicals	200.00	Required for experimental setup
Travel to Field Sites	150.00	To collect samples and conduct surveys
Data Analysis Software (SPSS)	100.00	For data entry and statistical analysis
Report Printing and Binding	50.00	For final report submission
Total Estimated Budget	800.00	—

When a justification is required, write a short paragraph explaining why each cost is essential to complete the research successfully.

Writing Tips

- Keep your budget realistic and transparent — no exaggerated or hidden costs.
- Match each cost item directly with your research activities.
- Include only necessary and justifiable expenses.
- Round off values logically and keep the total within feasible limits.
- If required, include a short justification (2–3 sentences per category).

5.12 Limitations of the Study (optional but useful)

Every research, no matter how well designed, has certain limitations. These are the factors that may affect the scope, accuracy, or generalization of the findings. Acknowledging them shows that the researcher is honest, realistic, and scientifically aware.

Purpose:

The purpose of this section is to show the boundaries of your study and to explain how these minor constraints will be managed or minimized. Mentioning limitations also helps readers interpret your results with a clear and fair understanding.

Common Types of Limitations

- Limited time or funding
- Small sample size
- Restricted access to data or respondents
- Minor errors in instruments or measurements
- Environmental or situational constraints

Example:

This study may face certain limitations such as limited time, budget, and sample size. These factors may slightly influence the generalization of results. However, all efforts will be made to minimize their effects through careful planning and consistent procedures.

Writing Tips

- Keep it short—one small paragraph is enough.
- Mention only real and relevant limitations.
- Use a positive tone—show awareness, not weakness.
- End with a note on how you will handle or reduce the limitations.

5.13 References/Bibliography & Citation Style

The References or Bibliography section lists all the sources that were cited or consulted during the preparation of your research proposal. It ensures academic honesty and allows readers to trace the original works you have used. Proper citation builds trust, credibility, and professionalism in your research.

Purpose: The purpose of referencing is to:

- Acknowledge the original authors or sources of information.
- Avoid plagiarism or copying without credit.
- Help readers find and verify the sources you used.
- Show that your study is based on authentic and scholarly evidence.

Citation Styles

Different disciplines use different citation formats. The most commonly accepted styles are:

Style	Commonly Used In	Example (In-text)	Example (Reference List)
APA (7th Edition)	Education, Social Science, Psychology	(Rahman, 2023)	Rahman, M. (2023). Research Methods in Education. Dhaka: Academic Press.
Harvard Style	General Sciences, Humanities	(Rahman 2023)	Rahman, M. 2023, Research Methods in Education, Academic Press, Dhaka.
IEEE Style	Engineering, Technology	[1]	[1] M. Rahman, Research Methods in Education, Academic Press, Dhaka, 2023.
Vancouver Style	Medical, Health Sciences	[1]	1. Rahman M. Research Methods in Education. Dhaka: Academic Press; 2023.

Basic Rules for Referencing

- All in-text citations must appear in the reference list.
- Arrange the reference list alphabetically by the author's last name.
- Use italic font for book and journal titles.
- Double-check the year, title, and page numbers for accuracy.
- Be consistent — use one citation style throughout your proposal.

Example (APA 7th Style)

In-text citation: "Immobilized microorganisms have been reported to improve removal efficiency (Wang et al., 2022)."

Reference list:

Wang, L., Chen, Y., & Rahman, M. (2022). Biological treatment of aquaculture wastewater using immobilized microbes. Journal of Environmental Science, 45(3), 120–128.

Bibliography (if required)

A Bibliography may include both cited and additional sources consulted for background reading.

It provides a broader list of materials that helped shape the research idea but were not directly quoted.

Writing Tips

- Choose one citation style (preferably APA 7th) and follow it consistently.
- Include at least 10–20 references for a strong proposal.
- Use recent and reliable sources (last 5–10 years).
- Always cite original sources instead of secondary ones.

5.14 Appendices (if needed)

The appendices include extra materials that support your research but are not essential in the main text. They help keep the proposal clear while providing full background details for readers who need them.

Purpose

To provide additional information, data, or documents that strengthen and validate your study. It shows that your research is complete, transparent, and well-documented.

What to Include

- Sample questionnaire or interview form
- Raw data or calculation tables
- Ethical approval or consent letter
- Maps, photos, or charts
- Experimental or analytical procedures
-

Format

- Label each as Appendix A, Appendix B, etc.
- Give a short title (e.g., Appendix A: Survey Questionnaire).
- Refer to them in the main text (e.g., "See Appendix A").
- Place each on a new page and keep formatting consistent.

Example:

- Appendix A: Sample Data Sheet
- Appendix B: Ethical Approval Form
- Appendix C: Experimental Setup Diagram

Appendices are the supporting evidence of your proposal —they add value and completeness without crowding the main report.

CHAPTER 6: FORMATTING, REFERENCING, & SUBMISSION

Once the content of your research proposal is complete, the final step is to format, reference, and submit it properly. This stage ensures that your proposal looks professional, follows academic standards, and is ready for official review. A well-formatted and properly referenced proposal reflects your discipline, attention to detail, and academic integrity.

6.1 Formatting the Proposal

Proper formatting gives your research a clear and organized appearance. It makes your proposal easy to read and professionally acceptable.

Basic Formatting Guidelines:

- Font: Times New Roman or Arial, size 12 pt
- Line spacing: 1.5 or double spacing
- Margins: 1 inch (2.54 cm) on all sides
- Page numbers: Bottom center or bottom right
- Paragraph alignment: Justified
- Headings and subheadings: Use consistent style (e.g., bold or numbered)
- Table and figure captions: Below figures and above tables, with numbering
- Paper size: A4, printed on one side only

Tip: Follow your university or funding agency's official formatting guide strictly.

6.2 Referencing and Citation Style

Referencing is a vital part of academic writing. It gives credit to other researchers and protects your work from plagiarism. Always use one consistent citation style throughout your proposal. Common Citation Styles:

- APA (7th Edition)—for social, biological, and environmental sciences
- Harvard Style—for general science and humanities
- IEEE Style—for engineering, computer, and applied sciences
- Vancouver Style—for medical and health sciences

Basic Rules:

- All in-text citations must appear in the reference list.
- Arrange references alphabetically (APA/Harvard) or numerically (IEEE/Vancouver).
- Include author(s), year, title, source, and page number (if available).
- Use recent and credible sources (preferably from the last 5–10 years).

Example (APA Style):

Rahman, M., & Chen, Y. (2023). Environmental Biotechnology and Wastewater Treatment. Journal of Applied Science, 12(4), 220–230.

6.3 Checking Before Submission

Before submitting your proposal, carefully review every section. Small mistakes can reduce the professional quality of your work. Checklist Before Submission:

- The title and abstract are clear and concise.
- All headings and subheadings are correctly numbered.
- Tables, figures, and references are properly formatted.
- Grammar, spelling, and punctuation are accurate.
- All citations are included in the reference list.
- The entire document follows the same formatting style.
- The proposal is within the page or word limit.

6.4 Submission Process

Follow your institution's or funding body's submission requirements carefully. Submission Options:

- Printed Copy: Submit one or more bound copies (spiral or soft cover).
- Digital Copy: Submit in PDF or Word format via email or online portal.
- Supporting Files: Attach appendices, ethical clearance, and consent forms if required.

Always meet the submission deadline and keep a backup copy of your final proposal.

6.5 Common Mistakes to Avoid

- Using mixed citation styles.
- Ignoring page layout and formatting rules.
- Submitting without proofreading.
- Missing references or incorrect numbering.
- Late submission or incomplete documentation.

CHAPTER 7: TIPS FOR A STRONG PROPOSAL PRESENTATION

Writing a research proposal is only one part of the process—presenting it effectively is equally important. A strong presentation helps convince reviewers that your research is clear, valuable, and feasible. It reflects not only your topic knowledge but also your communication and analytical skills.

7.1 Purpose of a Proposal Presentation

The main purpose of presenting your proposal is to:

- Explain your research idea clearly and confidently.
- Show your understanding of the topic and related literature.
- Defend your research design and methods.
- Receive constructive feedback before the full research begins.

7.2 Preparing for the Presentation

Preparation is the key to success. You must know your topic thoroughly and organize your slides or notes carefully. Steps to Prepare:

- Review your proposal and identify the core message.
- Prepare 10–15 slides (for a 10–12 minute presentation).
- Use simple, readable visuals — graphs, diagrams, key points.
- Practice explaining complex ideas in clear, short sentences.
- Time yourself to stay within the allowed duration.

Tip: Your goal is not to read the proposal but to tell the story of your research.

7.3 Structure of a Good Presentation

A strong research proposal presentation usually follows this order:

- Title Slide – Title, name, department, date.
- Background/Problem Statement—Why this study is important.
- Research Objectives/Questions—What you aim to find out.
- Literature Gap—What others have done and what is missing.
- Methodology—How you will conduct the research.
- Expected Results / Significance – What impact your study will create.
- Timeline and Budget (optional)—Work plan overview.
- Conclusion – Summarize and highlight research value.
- Q&A Session—Be ready for questions.
- Reference

7.4 Presentation Skills and Body Language

Your delivery style matters as much as your content. Speak with clarity, confidence, and enthusiasm.

Do's:

- Maintain eye contact with the audience.
- Speak slowly and clearly.
- Use confident posture and natural gestures.
- Emphasize key points with your tone.
- Smile and stay calm—even during tough questions.

Don’ts:

- Reading everything from the slides.
- Speaking too fast or too softly.
- Turning your back to the audience.
- Overloading slides with text.
- Showing nervous habits (fidgeting, unnecessary movements).

7.5 Visual and Technical Tips

- Use a clean, consistent slide design.
- Choose readable fonts (Arial, Calibri, Times New Roman).
- Use color sparingly and professionally.
- Include visuals (charts, diagrams) to simplify complex data.
- Keep each slide focused on one idea only.
- Always test your presentation equipment before starting.

7.6 Common Mistakes to Avoid

- Slides full of long paragraphs.
- Lack of eye contact or weak voice.
- Poor time management (too short or too long).
- Ignoring questions or reacting defensively.
- Overuse of animations or flashy graphics.

7.7 Final Tips for Success

- Know your research—be the expert of your topic.
- Keep your slides simple and your speech confident.
- Focus on clarity, logic, and enthusiasm.
- Rehearse multiple times before the actual presentation.
- End with a strong message—why your research matters.

A great proposal presentation is not about perfection — It’s about clarity, confidence, and conviction.

CONCLUSION

A research proposal is the beginning and foundation of any research work. It shows how a researcher plans, thinks, and prepares to find answers to important questions. Through this book, readers have learned how to choose a topic, identify a problem, set objectives, collect and analyze data, and finally prepare a clear proposal. Each step has been explained in a simple and practical way so that anyone can write a complete and meaningful research proposal.
In research, proper planning, time management, and honesty are essential.

Writing clearly, referencing correctly, and presenting confidently reflect the researcher's dedication and professionalism. A good proposal is more than just writing—it reflects a researcher's curiosity, creativity, and commitment to discovering new knowledge.

Good research begins with a good proposal, and a good proposal comes from clear thinking, careful planning, and sincere effort.

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